

SpotAI: Real vs AI Image Detector



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HOW DOES IT WORK?

2. IMAGE PREPROCESSING

All images are resized and normalized so they have the same format and quality. This ensures the model focuses on patterns in the image rather than differences in size or resolution.

4. MODEL TRAINING

The model makes predictions and compares them to the correct labels to calculate error.

It then adjusts its internal weights over multiple training cycles to improve accuracy.



1. DATA COLLECTION

The model is trained using a dataset of images labeled as either real or AI-generated. These examples allow the model to learn what visual differences exist between the two categories.

3. FEATURE LEARNING

A convolutional neural network analyzes images to learn visual features such as textures, edges, and inconsistencies. These features help the model recognize signs that an image was generated by AI.

5. IMAGE CLASSIFICATION

Once trained, the model can analyze new images it has never seen before. It outputs a prediction indicating whether the image is real or AI-generated, along with a confidence score.

WHICH IMAGE IS A.I. GENERATED?





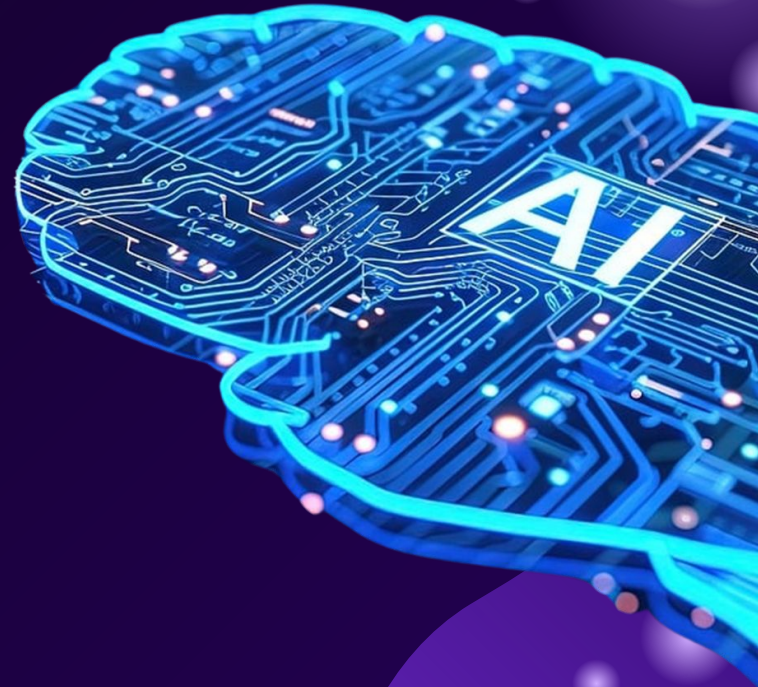
**WHICH IMAGE IS A.I.
GENERATED?**





As you saw

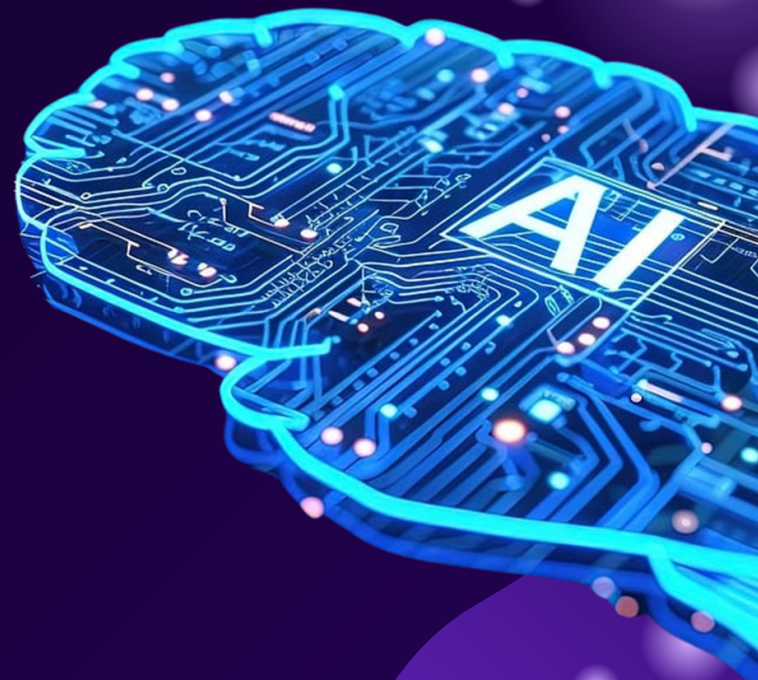
AI generated images are everywhere, and as time passes, it's getting harder and harder to tell the difference. Our solution? An AI analyzer to check images.



SpotAI

We created a simple machine-learning program called SpotAI that detects whether an image is AI-generated or real. We trained a model using labeled examples and then built a script that loads the trained model and evaluates new images. When we test an image, the program displays it on the screen along with a prediction and confidence score.

The goal was to make the process easy, fast, and understandable, showing how machine learning can identify subtle patterns that distinguish AI content from real photos.



STEP 1: DATA COLLECTION

Generative and machine learning models start by collecting large amounts of data. For SpotAI, this data consists of labeled images stored in folders.



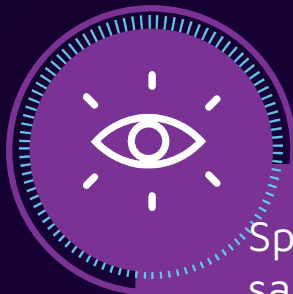
SpotAI collects images from a local dataset folder. Each image is automatically labeled based on its folder name (Real or AI), forming the foundation for training the model.

```
path = Path('data')  
path.ls()
```

```
images = get_image_files(path)  
print(f"Total images found: {len(images)}")
```

STEP 2: IMAGE PREPROCESSING

Before training, images must be standardized so the model can recognize visual patterns consistently.



SpotAI resizes all images to the same size and splits the data into training and validation sets. This helps the model focus on visual patterns instead of image size differences.

```
data_block = DataBlock(  
    blocks=(ImageBlock, CategoryBlock),  
    get_items=get_image_files,  
    splitter=RandomSplitter(valid_pct=0.2, seed=42),  
    get_y=parent_label,  
    item_tfms=Resize(224)  
)
```

STEP 3: FEATURE LEARNING

The model learns by analyzing patterns in the data and adjusting itself to reduce errors.



Our model uses a pre-trained ResNet18 neural network and fine-tunes it on our dataset for three training cycles (epochs). This allows the model to learn distinguishing features between the real and AI-generated images.

```
dls = data_block.dataloaders(path)
```

```
learn = vision_learner(dls, resnet18, metrics=accuracy)
```

STEP 4: MODEL TRAINING

During training, it makes predictions and compares them to the correct labels to calculate error (loss). It then updates its internal weights over multiple epochs to improve accuracy.

```
learn.fine_tune(3)
```



`fine_tune(3)` trains the pre-trained ResNet18 model for 3 epochs on our dataset. Fine-tuning adapts the model's learned knowledge to our specific detection task.

TRAINING RESULTS

epoch	train_loss	valid_loss	accuracy	time
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0	1.062106	0.710413	0.719388	00:25
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epoch	train_loss	valid_loss	accuracy	time
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0	0.724013	0.622980	0.704082	00:28
---	----------	----------	----------	-------

1	0.524566	0.749124	0.709184	00:28
---	----------	----------	----------	-------

2	0.412885	0.720685	0.729592	00:29
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Training Results

These results show how the model performed during training across several epochs. The table lists the model's training loss, validation loss, and accuracy after each round of learning. The training completed successfully, showing the model was able to learn patterns.

STEP 4: EVALUATION & IMAGE CLASSIFICATION

After training, the model is evaluated to understand its strengths and weaknesses.



The confusion matrix and classification report show how often SpotAI makes correct or incorrect predictions. This helps identify biases and areas for improvement.

```
interp = ClassificationInterpretation.from_learner(learn)
interp.plot_confusion_matrix()
```

```
def evaluate_model():
    preds, targs = learn.get_preds()
    print(classification_report(targs, preds.argmax(dim=1)))
```

```
evaluate_model()
```

RESULTS

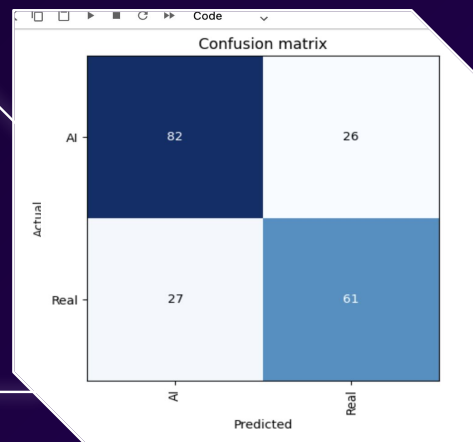
	precision	recall	f1-score	support
0	0.75	0.76	0.76	108
1	0.70	0.69	0.70	88
accuracy			0.73	196
macro avg	0.73	0.73	0.73	196
weighted avg	0.73	0.73	0.73	196

Classification Report

Overall, the classification report shows SpotAI learned to separate AI vs real images fairly well, with slightly stronger performance on the AI class. Most predictions are correct, but the model still makes some mix-ups because the two types of images can look very similar. With accuracy around the low-mid 70% range and balanced results across both classes.

Confusion Matrix

This confusion matrix shows how SpotAI performed on the validation dataset. The top-left and bottom-right cells represent correct predictions. SpotAI correctly identified 82 AI-generated images and 61 real images. Some errors occurred, where AI images were labeled as real and vice versa, which is expected with a limited dataset



STEP 5: CONTENT GENERATION

The trained model applies what it learned to new, unseen data.



SpotAI takes a new image and predicts whether it is Real or AI-generated. The result is displayed with a confidence score, demonstrating real-world usage of the model.

```
def show_and_predict(img_path):  
    pred, idx, probs = learn.predict(img_path)  
    confidence = probs[idx].item() * 100
```

```
show_and_predict('test_image2.png')
```

```
show_and_predict('test_image1.png')
```

RESULTS: Success!

Real (100.0% confidence)



Image: test_image2.png
Prediction: Real

AI (96.4% confidence)



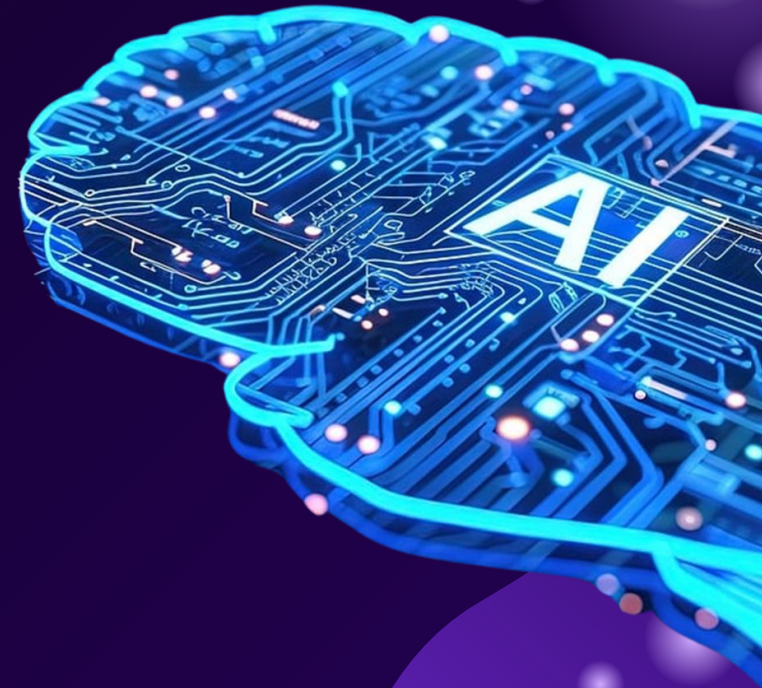
Image: test_image1.png
Prediction: AI

The model correctly identified the 2 photos each with high confidence. This shows that the model successfully learned the visual patterns and that it's a reliable AI image detector

Future Ideas

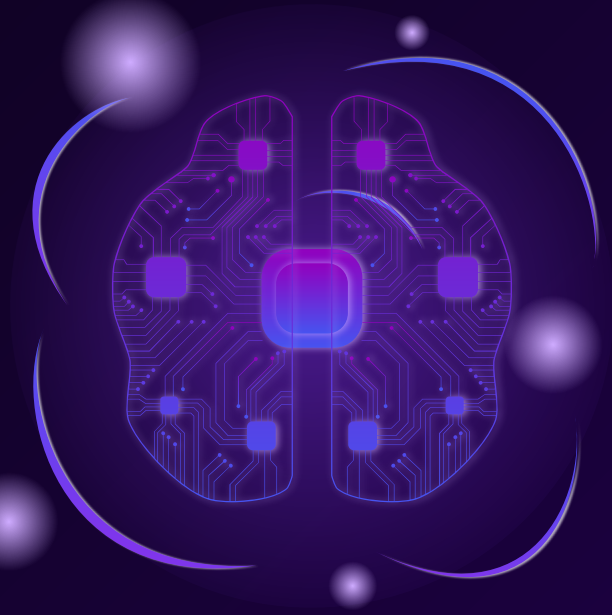
We plan on adding updates as we train it more to detect more complex images and videos as well. We also hope to add an update to check fake audios.

We also think it would be useful as an app or website so other people would be able to access whenever necessary



RESOURCES

- <https://teaching.pitt.edu/resources/what-is-generative-ai/>
- <https://www.kaggle.com/datasets/cashbowman/ai-generated-images-vs-real-images/code>
- <https://blog.qualys.com/product-tech/2025/04/18/risk-mitigation-for-generative-ai-workloads>
- <https://www.geeky-gadgets.com/generative-ai-societal-challenges/>
- <https://www.techtarget.com/searchenterpriseai/tip/Generative-AI-ethics-8-biggest-concerns>
- <https://www.youtube.com/watch?v=V9x6UyeknY4>



Thank You!